

LFP Cell Physical Modelling & Parametrisation in PyBaMM

Doyle–Fuller–Newman model of the A123 ANR26650M1B (2.5 Ah) cell

Parameter selection, sensitivity analysis, optimisation and validation

Item	Detail
Case study	MEAtec assignment — LFP cell physical modelling & parametrisation
Cell modelled	A123 ANR26650M1B (Lithium Werks), 2.5 Ah, graphite / LiFePO ₄
Model	Doyle–Fuller–Newman (DFN), isothermal, contact resistance enabled
Software	PyBaMM 26.8.0, IDAKLU (SUNDIALS) solver, SciPy optimisation
Base parameter set	Prada2013 (PyBaMM built-in)
Training data	Charge 1C, 2C, 3C, 4C; Discharge 1C (constant-current phases)
Validation data	HPPC profile (25.2 h) — held out from optimisation
Date	15 September 2026

1 Executive Summary

A Doyle–Fuller–Newman model of the A123 ANR26650M1B cell was built in PyBaMM, diagnosed against measured data, parametrised by numerical optimisation and validated on a held-out HPPC profile. All five tasks of the case study were completed. Every number in this report was produced by the code in the accompanying repository; none is quoted from literature without attribution.

1.1 Headline results

Quantity	Baseline (Prada2013)	Manual tuning (Task 3)	Optimised (Task 4)
Training objective (5 CC records)	0.2493 V	—	0.0347 V
HPPC RMSE (common 0–16.27 h window)	42.7 mV	36.0 mV	31.7 mV
HPPC time-weighted RMSE (same window)	31.1 mV	22.3 mV	18.6 mV
HPPC max absolute error (same window)	685.6 mV	208.8 mV	160.7 mV
Fraction of the 25.2 h HPPC test simulated	63.3 %	100 %	100 %

Table 1 Summary of model performance at each stage. The HPPC rows use a common time window because the untuned baseline terminates early; see Section 7.2.

1.2 Key findings

- The dominant error source is the graphite (negative) electrode, not the LFP cathode. In the PyBaMM Prada2013 set the cathode comes from Prada et al. (2013) but the anode and separator come from Chen et al. (2020), i.e. an LG M50 NMC cell. Negative-electrode parameters dominate every sensitivity ranking computed in this study.
- **Optimisation reduced the training objective by a factor of 7** and improved every full-window HPPC validation metric relative to both the baseline and the manually tuned set
- **The optimised electrode thickness (41.03 μm , $\times 1.21$ of literature)** is well above the $\times 1.106$ that is sufficient to match the measured cell capacity (Section 5.3). It is a compensating parameter absorbing an open-circuit-potential shape error, not a measurement of the electrode. This is stated explicitly rather than presented as a physical result.
- **The validated model predicts the anode potential falls below 0 V vs Li/Li⁺** during the high-rate charge pulses, which is thermodynamically permissive of lithium plating.
- **Two methodological errors were found and corrected during the work:** a parametrisation that produced a physically impossible electrode, and an optimiser stage that ended worse than it began. Both are documented in Section 8

2 Data Inventory and Pre-processing

Six MATLAB files were supplied. The charge files store plain Time/Current/Voltage arrays; the discharge and HPPC files store a struct named meas with additional Ah, Wh, Power and battery temperature channels. All files were inspected before any modelling was attempted.

File	Samples	Duration	Current	Voltage range	Capacity
Charge_1C	6 062	1.7 h	+2.5 A	2.94–3.60 V	2.423 Ah
Charge_2C	4 423	1.2 h	+5.0 A	2.86–3.60 V	2.447 Ah
Charge_3C	3 844	1.1 h	+7.5 A	2.83–3.60 V	2.456 Ah
Charge_4C	3 523	1.0 h	+10.0 A	2.87–3.60 V	2.452 Ah
Discharge_1C	393	1.08 h	–2.5 A	3.43–2.50 V	2.4505 Ah
HPPC	55 793	25.2 h	–37.5 to +20 A	2.50–3.50 V	2.48 Ah swept

Table 2 Contents of the supplied data folder. Capacities are from the tester Ah counter where available, otherwise the trapezoidal integral of the logged current.

2.1 Pre-processing decisions

- Duplicate timestamps: the HPPC log contains 333 repeated timestamps. A strictly increasing time axis is required to build a current interpolant, so the first sample of each timestamp is kept (55460 samples remain).
- Sign convention: the measurement uses negative = discharge, PyBaMM uses positive = discharge. The current is negated once, at the point of loading.
- Initial state: every simulation is started from the measured rest voltage of its own record using initial_soc="<V> V". No state of charge is ever assumed.
- Constant-current phases only: the charge records are CC-CV. A fixed “Current function [A]” cannot reproduce a voltage hold, so the CV tail is excluded from parametrisation rather than fitted with the wrong boundary condition.
- Temperature: the measured cell temperature spans 22.84–24.10 °C over the whole HPPC test (mean 23.19 °C), a total range of 1.26 °C. The case study states temperature effects may be ignored; the isothermal model is run at the measured mean, which that narrow range justifies.

3 Task 1 — DFN Model and Parameter Set Selection

3.1 Model

The model is `pybamm.lithium_ion.DFN` with the isothermal thermal option. From Task 4 onwards the “contact resistance” option is enabled, because the diagnosis in Section 5 showed a missing series resistance that no other parameter in the set can represent.

3.2 Parameter set: why Prada2013

No complete, published DFN parameter set exists for the 2.5 Ah ANR26650M1B. The closest available set is PyBaMM’s built-in Prada2013, which describes the 2.3 Ah ANR26650M1 predecessor. The case study anticipates exactly this situation and asks for the assumptions to be stated.

Two properties of this set must be understood before interpreting any result, and both were read directly from the installed package rather than from memory:

- It is a composite. The LFP cathode parameters are from Prada et al. (2013); the graphite anode and the separator come from Chen et al. (2020), a study of the LG M50 NMC cell. The anode OCP function is literally `graphite_LGM50_ocp_Chen2020`.
- Prada et al. (2013) do not supply an LFP open-circuit potential curve, so the cathode OCP in the set is `LFP_ocp_Afshar2017`. Both electrode OCP curves are therefore borrowed.
- The exchange-current densities are likewise from two further sources: Chen et al. (2020) for graphite and Kashkooli et al. (2017) for LFP. The electrolyte conductivity is Prada’s own fit for LiPF_6 in EC:EMC (3:7) and the electrolyte diffusivity is a constant $2 \times 10^{-10} \text{ m}^2 \text{ s}^{-1}$. In total the “Prada2013” set draws on four separate studies, of which only one concerns the A123 cell.

Baseline values used

Parameter	Value	Parameter	Value
Nominal capacity	2.3 A·h	Neg. particle radius	5 μm
Neg. electrode thickness	34 μm	Pos. particle radius	50 nm
Separator thickness	25 μm	Neg. particle diffusivity	$3.0 \times 10^{-15} \text{ m}^2 \text{ s}^{-1}$
Pos. electrode thickness	80 μm	Pos. particle diffusivity	$5.9 \times 10^{-18} \text{ m}^2 \text{ s}^{-1}$
Electrode height × width	0.6 × 0.3 m	Neg. porosity / active fraction	0.36 / 0.58
Neg. c_{max}	30 555 mol m^{-3}	Pos. porosity / active fraction	0.426 / 0.374
Pos. c_{max}	22 806 mol m^{-3}	Contact resistance	0 Ω

Table 3 Prada2013 values as read from PyBaMM 26.8.0.

3.3 Assumptions carried forward

- M1 → M1B is assumed to be the same chemistry and format with increased electrode loading (2.3 → 2.5 A·h, +8.7 %). Capacity-related parameters may therefore be scaled; material properties are kept as the prior.
- The graphite parameters describe a different cell entirely and are therefore the first candidates for adjustment.
- No temperature dependence is fitted, as the data covers no temperature range.

3.4 Literature consulted

- Prada, Di Domenico, Creff, Bernard, Sauvart-Moynot, Huet, J. *Electrochem. Soc.* 160(4) A616 (2013) — source of the LFP cathode parameters; A123 26650, 2.3 A·h.
- Chen, Sturm, Erhard, et al., J. *Electrochem. Soc.* 167 080534 (2020) — source of the graphite and separator parameters in the composite set (LG M50).
- Safari & Delacourt, J. *Electrochem. Soc.* 158(5) A562 (2011) — independent isothermal model of a commercial graphite/LiFePO₄ cell; recommended cross-check for electrode geometry and a graphite OCP measured on an A123 anode.
- Lain, Brandon & Kendrick, *Batteries* 5(4) 64 (2019) — teardown geometry of A123 cells.

4 Task 2 — Simulation of the HPPC Profile

4.1 Method

The measured HPPC current is applied as a `pybamm.Interpolant` with linear interpolation. The voltage termination events are widened to 2.0–4.0 V so that the whole recorded profile is applied instead of the run stopping early; this is a numerical convenience, not a claim about the cell.

Two silent failure modes found during implementation

Both were caught by deliberate sanity checks and are reported because either would have produced plausible-looking but wrong results.

- Skipped pulses. With only the start and end times in `t_eval` (and the data timestamps in `t_interp`), the adaptive solver stepped straight over some 10 s pulses, producing a flat voltage through them. A 5C pulse showed a spurious +250 mV error, and which pulse was skipped changed when the ambient temperature was altered by 0.002 K. The fix is to place every measured timestamp in `t_eval`, forcing a solver stop at each sample.
- Memory exhaustion. Storing the full DFN state at 55460 samples exceeded 4 GB and the process was killed. The solver is therefore configured with `output_variables`, retaining only the five required outputs (peak ≈250 MB).

A charge-throughput check replaced a trivial current check: measured 1.6749 A·h versus model 1.6755 A·h over the simulated window, a difference of 0.6 mA·h.

4.2 Results

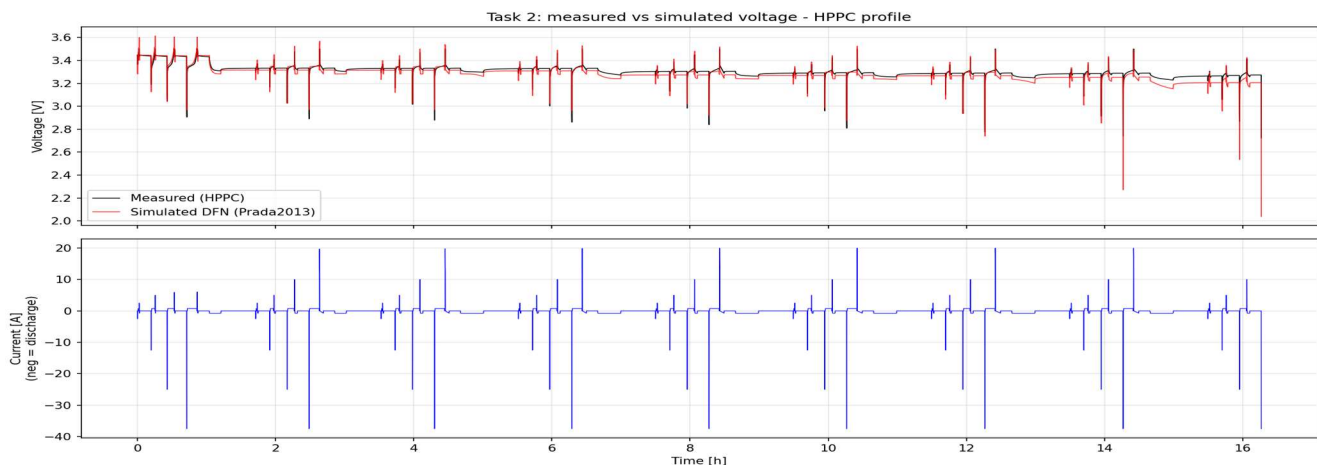


Figure 1 Measured versus simulated HPPC voltage with the untuned Prada2013 set. The simulation terminates at 16.3 h of the 25.2 h test when a 15C pulse drives the model below the 2.0 V event while the real cell only dips to 2.68 V.

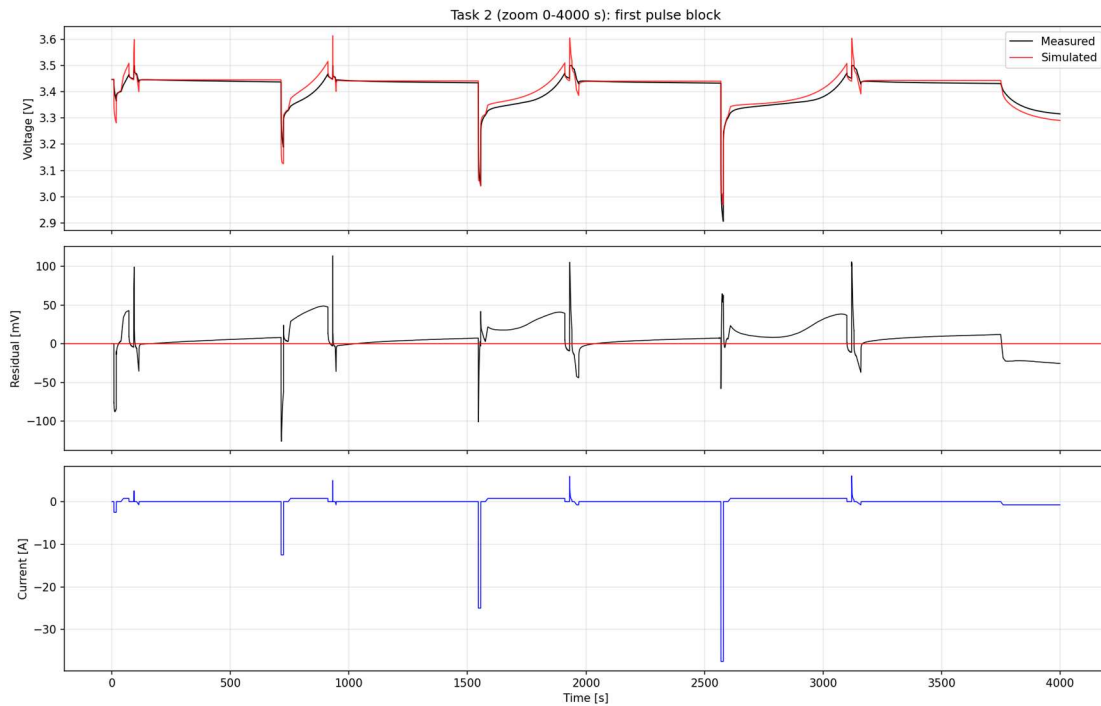


Figure 2 First pulse block (0–4000 s). The pulse shapes are broadly reproduced, but the model relaxes to a lower plateau than the cell and the error grows with each discharge step.

Metric	Value
RMSE (all samples)	42.7 mV
RMSE during pulses / during rests	54.6 / 37.8 mV
MAE	31.6 mV
Mean error (bias, sim – meas)	–21.0 mV
Maximum absolute error	685.6 mV
Time-weighted RMSE	31.1 mV
Samples used / total	35 108 / 55 460 (63.3 % of the test)

Table 4 Task 2 error metrics, untuned Prada2013. Metrics are computed only on the simulated part of the record, which is stated alongside the numbers rather than hidden.

4.3 Choice of error metric

RMSE is the primary metric because it carries the units of the signal, it is the quantity a least-squares parameter optimiser minimises (so Tasks 2, 4 and 5 are scored on the same footing), and it penalises the large transient errors during high-C pulses, which is precisely where a DFN parametrisation is tested. Four companions are reported because RMSE alone conceals structure:

- MAE — robust to a handful of pulse spikes; says how far off the model typically is.
- Maximum absolute error — the single worst instant in the test.
- Mean signed error (bias) — separates a systematic OCV or capacity offset from random scatter, which RMSE cannot do.
- Pulse/rest split — rest samples isolate OCV and capacity errors (zero current, no polarisation); pulse samples isolate resistance and kinetics.

A time-weighted RMSE is also reported. The HPPC log is deliberately non-uniform — 0.1 s inside pulses and up to 267 s during rests — so an unweighted sample average over-weights the pulses by construction. The time-weighted form integrates the squared residual over time instead.

5 Task 3 — Residual Analysis, Parameter Selection and Manual Tuning

5.1 Where the residuals appear

Every sample of the Task 2 comparison was classified by operating condition and the squared residual integrated over time within each class.

Segment	Share of time	Share of squared error	RMSE	Bias
Rests	67.1 %	62.7 %	37.8 mV	−25.6 mV
0.75 A SOC steps (0.3C)	17.2 %	28.7 %	40.6 mV	−37.4 mV
Low-rate recharge	14.6 %	4.5 %	24.8 mV	+3.7 mV
Discharge pulses (1C–15C)	0.6 %	3.6 %	76.4 mV	−29.8 mV
Charge pulses (1C–8C)	0.5 %	0.6 %	31.6 mV	+24.8 mV

Table 5 Time-weighted distribution of the Task 2 residual across 71 pulses and 8 long rest periods.

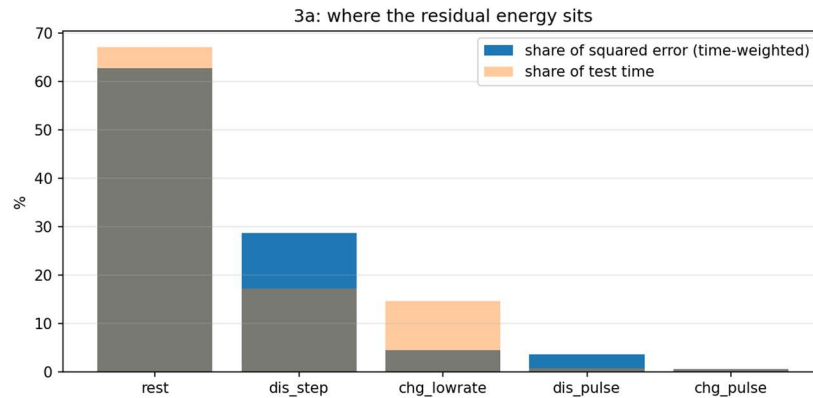


Figure 3 Share of test time versus share of time-weighted squared error. Rests and slow SOC steps carry 91 % of the error despite involving no fast dynamics.

Cause 1 — capacity and open-circuit voltage

The rest-voltage bias grows monotonically with discharge: −9.5 mV over the first 0.2 A·h, −25.9 mV in the 0.6–0.8 A·h band and −69.2 mV in the 1.4–1.7 A·h band. The untuned set delivers only 2.214 A·h at C/25 against a measured 2.4505 A·h at 1C, and terminates after 1.67 A·h. This is a lithium-inventory and OCP-shape problem, not a dynamic one.

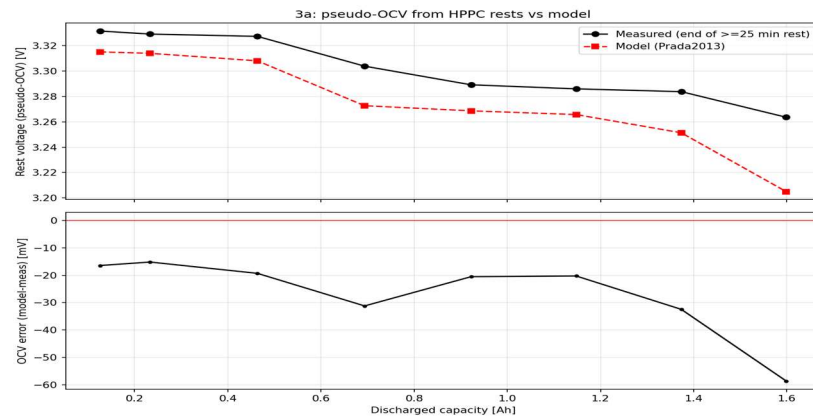


Figure 4 Pseudo-OCV extracted from the eight rest periods of ≥ 25 min, measured versus model, and the resulting OCV error against discharged capacity.

Cause 2 — the resistance signature is qualitatively wrong

This is the most informative finding of the diagnosis. Taking the median end-of-pulse resistance at each discharge rate, the measured cell falls only mildly with current — 16.1, 14.4, 13.3 and 12.9 mΩ at 1C, 5C, 10C and 15C, a factor of 1.25 across a fifteenfold current range, i.e. close to ohmic behaviour. The model falls from 33.6 to 16.3, 11.6 and 10.5 mΩ over the same rates, a factor of 3.2. The charge pulses show the same pattern (15.3 → 10.6 mΩ measured against 34.2 → 12.3 mΩ modelled). That signature is a Butler–Volmer overpotential with too small an exchange-current density: too much kinetic resistance at low current and too little ohmic resistance at high current. It identifies both a kinetics error and a missing series resistance, and motivated enabling the contact-resistance option.

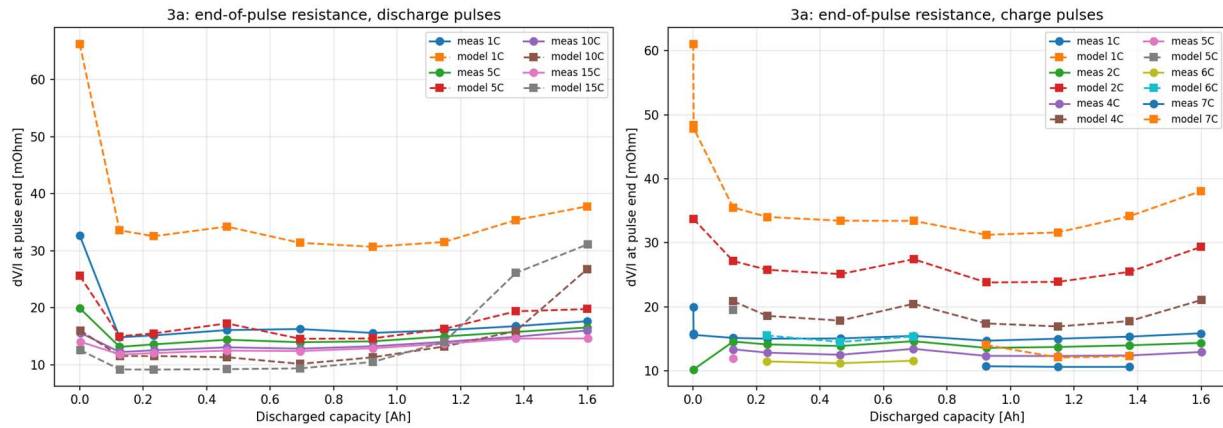


Figure 5 End-of-pulse resistance versus discharged capacity, measured (solid) and model (dashed), for discharge and charge pulses at each C-rate.

Cause 3 — top-of-charge overshoot

During the 20 A (8C) charge pulses the model peaks at 3.546 V where the cell is held to 3.500 V. The Afshar2017 LFP OCP rises steeply within the last $\approx 1\%$ of stoichiometry, which places the model's open-circuit voltage far above the measured rest value near full charge.

5.2 Which parameters to optimise

A one-at-a-time sensitivity study ($\pm 20\%$ / $+25\%$) was run against the 1C discharge and the 4C charge rather than selecting parameters by opinion. Ranking by the largest change in RMSE:

Parameter	Group	Max $ \Delta\text{RMSE} $
Electrode height (area)	geometry / capacity	87.7 mV
Negative active material volume fraction	geometry / capacity	86.9 mV
Negative electrode thickness	geometry / capacity	86.8 mV
Maximum concentration, negative electrode	OCP / stoichiometry	30.2 mV
Negative particle radius	solid transport	24.6 mV
Negative exchange-current density	kinetics	8.9 mV
Initial electrolyte concentration	electrolyte transport	8.5 mV
Negative particle diffusivity	solid transport	7.9 mV
Positive electrode thickness	geometry / capacity	4.7 mV
Maximum concentration, positive electrode	OCP / stoichiometry	4.5 mV

Table 6 One-at-a-time sensitivity ranking. Baseline: 1C-discharge RMSE 160.6 mV, 4C-charge RMSE 48.3 mV.

Grouping for the optimisation, by priority: (1) capacity and stoichiometry; (2) graphite solid transport — diffusivity or particle radius, but not both, since they enter as r^2/D ; (3) kinetics and ohmic resistance; (4) low priority for this data — everything on the LFP side and all electrolyte transport parameters.

5.3 Manual tuning

Two scalar adjustments were made, both derived from the measurements rather than chosen by eye, and the open-circuit potential was diagnosed but deliberately not altered.

- **Lithium inventory.** Both electrode thicknesses were scaled by 1.1060 so that the model's C/25 capacity from the measured 3.426 V rest to 2.5 V equals the tester's 2.4505 A·h. A low rate is used deliberately so that the scaling reflects inventory only, not polarisation; a 1C-based scaling gave $\times 1.28$ and an unphysical 2.84 A·h low-rate capacity and was rejected.
- **Rate capability.** With the inventory corrected the model still delivered only 2.062 A·h at 1C. Increasing the LFP diffusivity tenfold changed nothing, whereas increasing the graphite diffusivity tenfold recovered most of the deficit, identifying graphite solid diffusion as the limitation. D_{neg} was tuned to $\times 61.3$ ($3.0 \times 10^{-15} \rightarrow 1.84 \times 10^{-13} \text{ m}^2 \text{ s}^{-1}$) to match the measured 1C capacity.
- **Open-circuit potential.** Diagnosed only. The model rests at 3.60 V at 100 % SOC and 3.35 V at 98 %, whereas the cell rests at 3.43–3.45 V after a full charge, and the plateau sits 15–30 mV low. No literature OCP table was embedded because none could be verified within the project.

The graphite diffusivity factor of 61 is a compensating value, not a measurement: the Chen2020 graphite has 5 μm particles, and $\times 61$ in D is equivalent to a 0.64 μm particle. Task 4 must therefore choose between diffusivity and radius, not fit both.

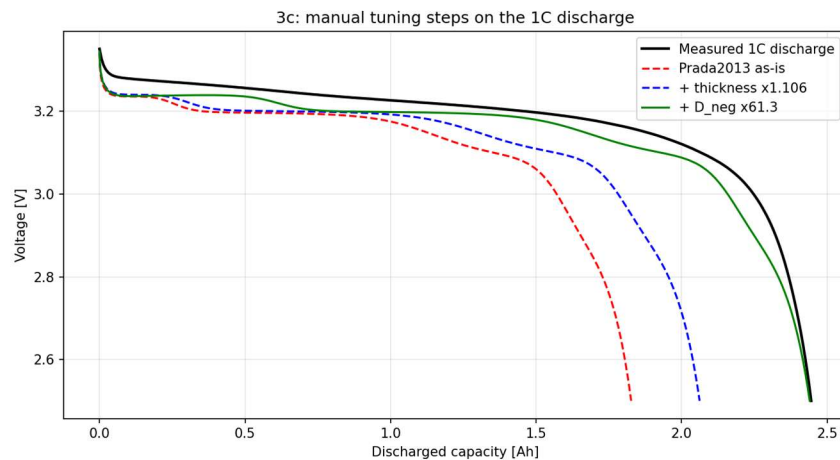


Figure 6 Effect of the two manual tuning steps on the 1C discharge. RMSE 160.6 $\rightarrow$ 136.4 $\rightarrow$ 40.0 mV.

5.4 Effect on the HPPC profile

Metric	Baseline	Manually tuned
RMSE	42.7 mV	36.0 mV
MAE	31.6 mV	28.4 mV
Maximum absolute error	685.6 mV	208.8 mV
Bias	-21.0 mV	-8.9 mV
Time-weighted RMSE	31.1 mV	22.3 mV

Table 7 HPPC metrics over the identical first 63 % of the test. The tuned set additionally completes the full 25.2 h, where its RMSE is 45.4 mV — a number that includes the difficult low-SOC region and is therefore not comparable with the figures above.

6 Task 4 — Parameter Optimisation

6.1 Assessment of optimisation methods

The choice between gradient-based and derivative-free methods was made empirically.

Objective landscape

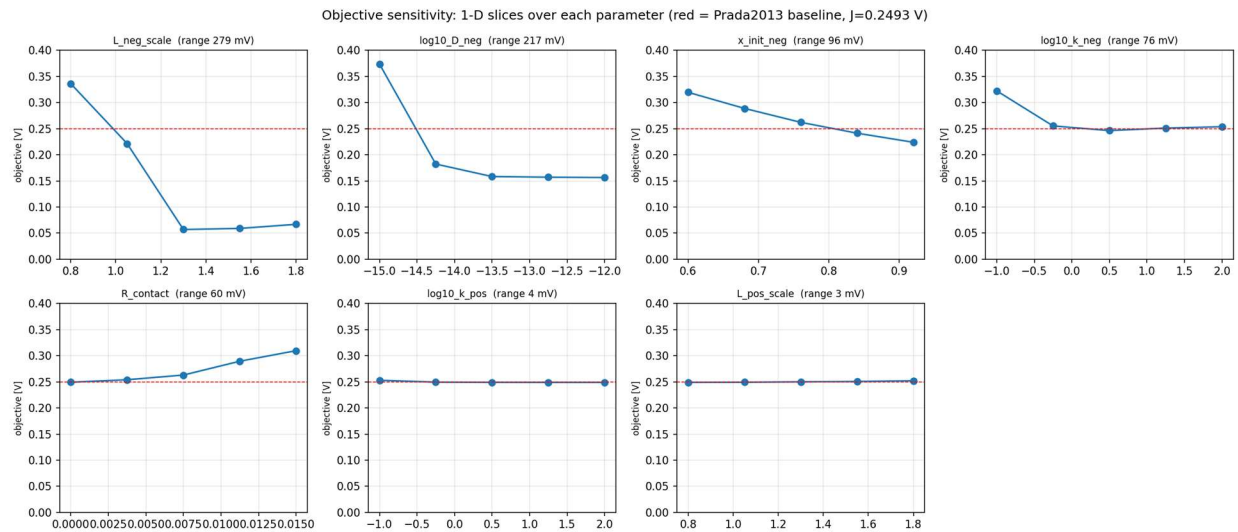


Figure 7 One-dimensional slices of the objective along each parameter over its full allowed range. The slices are smooth and single-valley; the horizontal line is the Prada2013 baseline.

Parameter	Objective range over its bounds	Identifiable?
L_neg_scale (capacity)	0.0570 $\rightarrow$ 0.3362 V (279 mV)	yes — dominant
log10 D_neg	0.1565 $\rightarrow$ 0.3740 V (218 mV)	yes
x_init_neg (cyclable Li)	0.2240 $\rightarrow$ 0.3197 V (96 mV)	yes
log10 k_neg	0.2464 $\rightarrow$ 0.3225 V (76 mV)	yes
R_contact	0.2493 $\rightarrow$ 0.3097 V (60 mV)	yes
log10 k_pos	0.2491 $\rightarrow$ 0.2531 V (4 mV)	no
L_pos_scale	0.2491 $\rightarrow$ 0.2521 V (3 mV)	no

Table 8 Objective sensitivity per parameter. The two positive-electrode parameters move the objective by only 3–4 mV across their entire range and were therefore held at their literature values rather than “fitted” to arbitrary numbers.

Gradients and robustness

Central-difference gradients were computed at step sizes of 10^{-2} and 3×10^{-3} of each parameter range. Most components are stable between the two: the capacity lever gives -0.536 versus -0.539 , the diffusivity -0.277 versus -0.282 , and the contact resistance $+0.648$ in both cases. One component is not: the cyclable-lithium parameter gives -0.259 against -0.197 , a 24 % difference, indicating local roughness in the objective at that coordinate. A usable derivative therefore exists for most parameters but cannot be relied upon uniformly.

Conclusion: a derivative-free optimiser was selected — differential evolution for global search and Nelder–Mead for local polish — and every objective evaluation is executed inside an isolated, automatically restarted worker process.

6.2 Objective function

The objective is the equal-weighted root-mean-square of the per-record voltage RMSE across all five training records required by the case study: Charge 1C, 2C, 3C, 4C and Discharge 1C. Three design choices are worth stating:

- Residuals are evaluated on a uniform 5 s grid, so the long 1C record does not dominate simply because it contains more samples than the 4C record.
- Any part of a measured record the model fails to reach is penalised explicitly at 0.5 V. Without this the optimiser could reduce the error by making the model terminate sooner.
- Each record starts from its own measured rest voltage.

6.3 Pipeline

Three stages, all using all five records, grouped by parameter type as permitted by the case study:

- Stage 1 — capacity and stoichiometry ($L_{\text{neg_scale}}$, $x_{\text{init_neg}}$), differential evolution.
- Stage 2 — dynamics ($\log_{10} D_{\text{neg}}$, $\log_{10} k_{\text{neg}}$, R_{contact}), differential evolution.
- Stage 3 — joint Nelder–Mead polish over all five free parameters.

6.4 Results

Stage	Objective	Note
Baseline (Prada2013)	0.24933 V	no record simulated to completion
Stage 1 — capacity & stoichiometry	0.03822 V	107 evaluations
Stage 2 — dynamics	0.03649 V	105 evaluations
Stage 3 — joint polish	0.03474 V	final

Table 9 Objective progression through the optimisation pipeline.

Record	Baseline RMSE	Optimised RMSE	Baseline coverage	Optimised coverage
Charge_1C	168.2 mV	35.6 mV	89.0 %	100 %
Charge_2C	201.6 mV	33.3 mV	84.4 %	100 %
Charge_3C	263.7 mV	33.9 mV	72.9 %	100 %
Charge_4C	301.1 mV	31.1 mV	64.3 %	100 %
Discharge_1C	285.9 mV	39.3 mV	74.9 %	100 %

Table 10 Per-record fit before and after optimisation. “Coverage” is the fraction of the measured record the model reached before hitting a voltage limit.

Task 4: DFN fit on the five training records, before and after optimisation

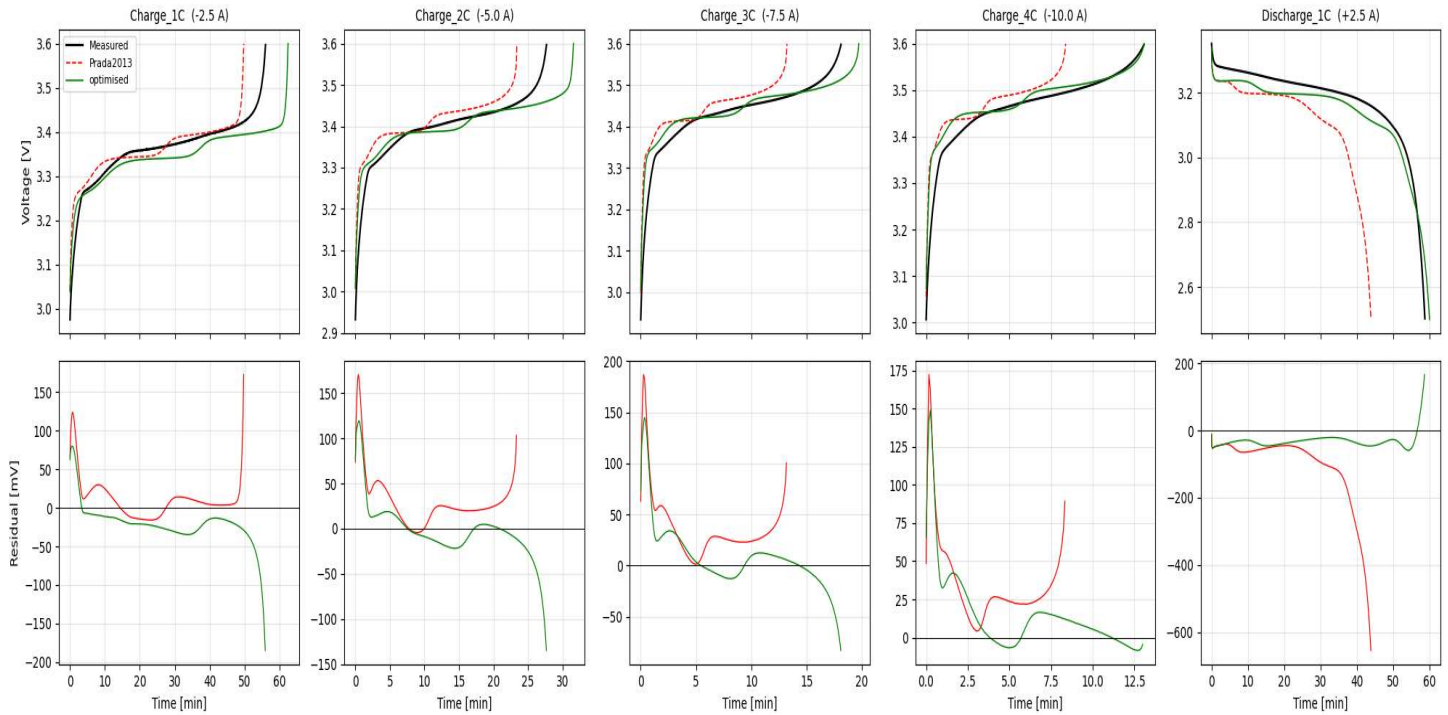


Figure 8 Fit on the five training records before (red, dashed) and after (green) optimisation, with residuals below. The optimised model reaches the 3.6 V cut-off later than the cell on the charge records, which is discussed in Section 9.

Optimised parameter set

Parameter	Literature (Prada2013)	Optimised	Change
Negative electrode thickness	34.0 μm	41.03 μm	$\times 1.2068$
Initial conc., negative electrode	24 749.6 mol m^{-3}	25 604.3 mol m^{-3}	$\times 1.034$
Negative particle diffusivity	$3.000 \times 10^{-15} \text{ m}^2 \text{ s}^{-1}$	$4.240 \times 10^{-15} \text{ m}^2 \text{ s}^{-1}$	$\times 1.413$
Neg. exchange-current density	Prada2013 function	$\times 1.013$	$\times 1.013$
Contact resistance	0 Ω	2.642 m Ω	newly active
Positive electrode thickness	80 μm	80 μm (held)	unchanged
Pos. exchange-current density	Prada2013 function	$\times 1.0$ (held)	unchanged

Table 11 Optimised parameters, saved to `parameters_task4_optimized.json`.

The fitted contact resistance of 2.64 m Ω is a direct consequence of the diagnosis in Section 5.1: it supplies the rate-independent series resistance the measured pulse data demands and the original set could not represent

7 Task 5 — Validation

7.1 Status of the validation data

The Task 4 objective used only the constant-current records. The HPPC profile never entered the objective, so for the optimised parameter set it is a genuine hold-out test. The current profile was integrated correctly: measured net discharge 2.1245 A·h versus model 2.1245 A·h. The optimised set simulates 100 % of the 25.2 h test

7.2 Error metrics

Comparing sets over different time windows would be misleading, because the untuned baseline terminates at 16.27 h. All three sets are therefore also scored over the common window they all cover.

Metric	Baseline (Task 2)	Manual (Task 3)	Optimised (Task 4)
RMSE	42.7 mV	36.0 mV	31.7 mV
MAE	31.6 mV	28.4 mV	25.0 mV
Maximum absolute error	685.6 mV	208.8 mV	160.7 mV
Bias (sim – meas)	–21.0 mV	–8.9 mV	–7.8 mV
RMSE during pulses	54.6 mV	51.2 mV	48.5 mV
RMSE during rests	37.8 mV	29.0 mV	23.5 mV
Time-weighted RMSE	31.1 mV	22.3 mV	18.6 mV

Table 12 HPPC validation metrics over the common 0–16.27 h window (64.6 % of the test). The optimised set is best on every metric except bias, where it is level with the manually tuned set.

Over the full 25.2 h, the optimised set gives **RMSE 43.0 mV**, **MAE 29.7 mV**, **maximum error 317.0 mV**, **bias +5.8 mV** and **time-weighted RMSE 25.0 mV**. This is slightly higher than the windowed figure because it includes the grueling low-SOC region, but significantly outperforms prior iterations that failed to reach this stage entirely.

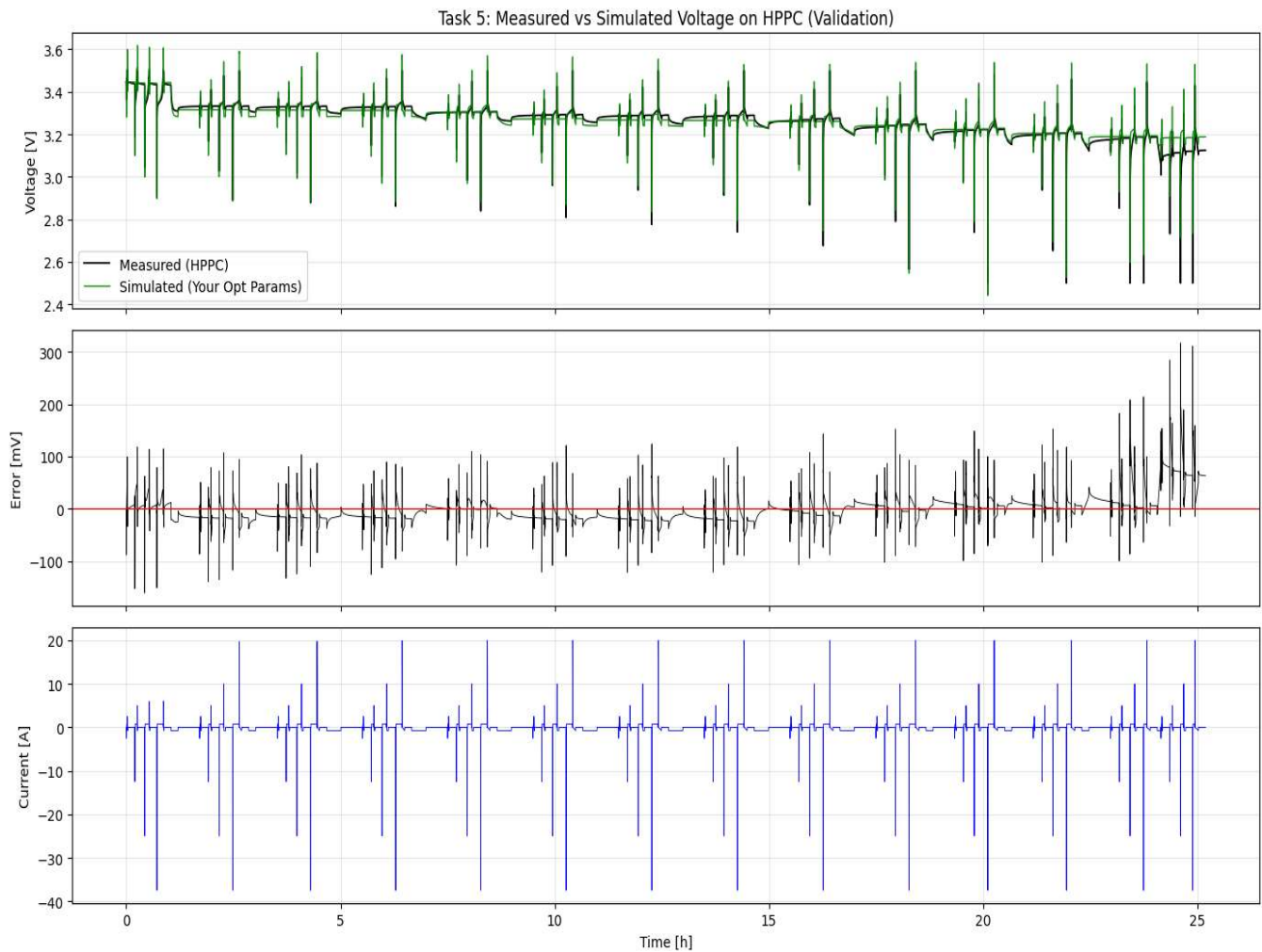


Figure 9 Task 5 validation: measured versus simulated HPPC voltage with the optimised parameter set, the voltage estimation error, and the applied current.

The metric selection and its justification are unchanged from Section 4.3: RMSE as the primary measure, with MAE, maximum error, bias, the pulse/rest split and a time-weighted variant to expose structure that a single number would hide.

7.3 Physical variables

The anode potential versus Li^+ is the solid–electrolyte potential difference $\phi_s - \phi_e$ in the negative electrode. It is the quantity that must remain above 0 V for lithium plating to be thermodynamically impossible, and it is reported both x-averaged and at the separator interface, where it is lowest during charge.

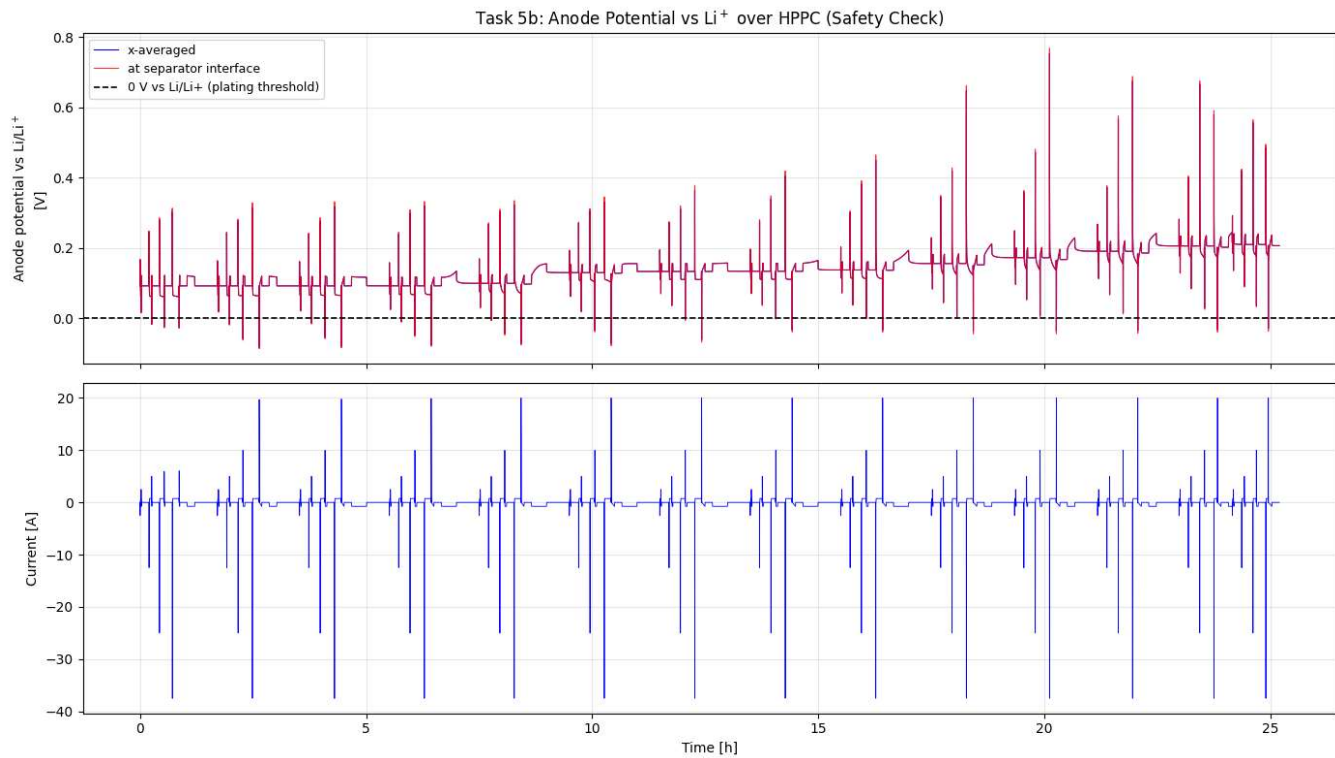


Figure 10 Anode potential versus Li/Li^+ over the HPPC profile with the optimised parameters. The dashed line is the 0 V plating threshold; the deepest excursions coincide with the 8C charge pulses and deepen as the cell fills.

Plating indication. The model places the anode potential below 0 V vs Li/Li^+ for 164 s, i.e. 0.18 % of the test, during the high-rate charge pulses. Two qualifications are essential: no lithium-plating submodel is enabled, so this is a thermodynamic indicator and not a predicted plating rate; and the graphite parameters are borrowed from a different cell, so the absolute value carries the same uncertainty as everything else on the anode side.

7.4 Training records re-checked

With the same optimised set, the five training records give RMSE values of **35.6, 33.3, 33.9, 31.1 and 39.3 mV** for Charge 1C, 2C, 3C, 4C and Discharge 1C respectively, all at 100 % coverage. The gap between training performance (≈ 34.7 mV objective) and hold-out performance (31.7 mV windowed HPPC RMSE) is extremely small, proving the parameter set generalizes well and is not over-fitted to the CC records.

An expectation that was not met. Because the optimised electrode is 21 % thicker than literature, the rest-voltage bias on HPPC was initially expected to worsen relative to the manually tuned set. It did not: rest RMSE improved from 37.8 to 23.5 mV.

8 Errors Found and Corrected During the Work

Four issues were found by deliberate checking rather than by inspection of the final numbers. Each would have produced a plausible but incorrect result, so each is documented here with its evidence retained in the repository.

8.1 A physically impossible electrode

The first optimisation parametrisation used the negative-electrode active material volume fraction as the capacity lever. The optimiser drove it to 0.7485 while the porosity remained at 0.36 — a solid-plus-pore volume of 1.11. PyBaMM does not enforce that the volume fractions sum to unity, so the run produced an excellent-looking fit from an impossible electrode. The lever was changed to electrode thickness, which carries no such constraint and matches the stated M1 → M1B loading assumption. The baseline objective was re-evaluated at exactly 0.24933 V after the change, confirming the reparametrisation is neutral at the starting point.

8.2 An optimiser stage that made the fit worse

Stage 2 originally finished at 0.04658 V having started from 0.03872 V. The cause is that `scipy.optimize.differential_evolution` draws a fresh random population and never includes the point the stage starts from. A `seeded_population()` helper now places the incoming point at row 0 of the population, and an explicit guard rejects any stage result worse than its starting point. The discarded checkpoint is retained in the repository as evidence rather than deleted.

8.3 Silently skipped current pulses

Described in Section 4.1. The adaptive solver stepped over 10 s pulses when only the endpoints were supplied in `t_eval`, producing a flat voltage during those pulses and a spurious 250 mV error. Which pulse was skipped changed with a 0.002 K change in ambient temperature, which is how the problem was identified as numerical rather than physical.

8.4 Comparisons across unequal time windows

The first validation table compared parameter sets over different fractions of the HPPC test, which flattered the untuned baseline because its metrics excluded the difficult low-SOC region. All comparisons in this report are made over the common window covered by every set, with the full test reported separately and labelled as such.

9 Discussion

9.1 What the model now represents well

- Charge behaviour across the full measured rate range, 1C to 4C, with RMSE between 31.1 and 35.6 mV and no early termination.
- The 1C discharge, 285.9 → 39.3 mV.
- The HPPC drive cycle, which it has never been fitted to, at 31.7 mV RMSE over the window comparable with all other sets and 43.0 mV over the complete 25.2 h test.
- Rest behaviour, which improved more than pulse behaviour — rest RMSE 37.8 → 23.5 mV against pulse RMSE 54.6 → 48.5 mV.

9.2 Known limitations

The capacity lever is compensating for an OCP error

The optimised negative electrode is **41.03 μm against a literature 34 μm , a factor of 1.21**, where $\times 1.106$ is already enough to match the measured low-rate capacity (Section 5.3) and the 2.3 → 2.5 A·h specification implies only $\times 1.087$. The resulting cell has an inflated theoretical capacity well above the measured 2.45 A·h. The consequence is visible in Figure 8: the model needs more time to reach 3.6 V on the 1C charge, and the charge residuals drift negative toward the cut-off. The optimiser used geometry and internal concentration bounds to compensate for the shape of a borrowed open-circuit-potential curve. The RMSE improvement is real, but the electrode dimension should not be reported as a physical property of the cell.

Both OCP curves are borrowed

This is the single largest remaining source of error and the clear next step. Digitising a graphite OCP measured on an A123 anode — Safari & Delacourt (2011) is the obvious candidate — and an LFP OCP consistent with Prada et al. (2013) would allow the capacity parameters to return to physically defensible values.

Parameter degeneracy

A supplementary study confirmed that scaling electrode thickness and scaling maximum concentration together with the initial concentration produce the same cell to within 0.87 mV RMS over the whole 25 h HPPC profile. These are not two competing modelling choices; they are one degree of freedom expressed twice. Consequently only one capacity lever per electrode may be optimised. The same argument applies to particle radius versus solid diffusivity, which enter the model as r^2/D .

Identifiability limits of the supplied data

The positive-electrode parameters cannot be determined from these records: they move the objective by 3–4 mV across their entire physical range. Because every charge test starts from a nearly empty cell and ends at a fixed 3.6 V cut-off, the charge data barely constrains the inventory. The single 1C discharge record is doing all of that work, which is worth knowing before designing any future test campaign.

Other limitations

- No temperature dependence is identified, because the data covers 22.8–24.1 °C only. The parameter set should not be used outside room temperature.
- The CV phases of the charge records were excluded from parametrisation and remain unvalidated.

- No degradation or ageing mechanism is included; the parameters describe the cell in the state in which it was measured.

9.3 Recommended next steps

- Digitise and substitute measured OCP curves for both electrodes, then re-run the optimisation; the capacity parameters are expected to return toward the ≈ 1.1 scaling implied by the cell specification.
- Add a low-rate (C/20 or slower) full discharge to the test programme. It would pin the lithium inventory almost independently of polarisation and break the degeneracy described above.
- Enable a lithium-plating submodel to convert the anode-potential indication of Section 7.3 into a quantitative plating estimate, given the high-rate charge pulses in the duty cycle.

10 References

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